Complex Numbers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The complex conjugate and division**

Find the conjugate of $z = 7 - 2i$, and compute $z\bar{z}$.

$\bar{z} = 7 + 2i$. $z\bar{z} = (7 - 2i)(7 + 2i) = 49 - 4i^2 = 49 + 4 = 53$.

2 **[Foundational]** • **Target: The complex conjugate and division**

Simplify $\frac{3+i}{2-3i}$ by multiplying by the conjugate of the denominator.

$\frac{3+i}{2-3i} \cdot \frac{2+3i}{2+3i} = \frac{(3+i)(2+3i)}{4+9} = \frac{6+9i+2i+3i^2}{13} = \frac{3+11i}{13} = \frac{3}{13} + \frac{11}{13}i$.

3 **[Foundational]** • **Target: Modulus and the Argand diagram**

Find the modulus $|z|$ of $z = 5 - 12i$.

$|z| = \sqrt{5^2 + (-12)^2} = \sqrt{25 + 144} = \sqrt{169} = 13$.

4 **[Foundational]** • **Target: Modulus and the Argand diagram**

The graph shows several complex numbers plotted on the Argand diagram (real part horizontal, imaginary part vertical).

a Read off the complex number plotted at $(4, 1)$ in Cartesian form. The point $(4, 1)$ corresponds to $4 + i$.

b Find the modulus of the complex number plotted at $(-2, -3)$. The point $(-2, -3)$ corresponds to $-2 - 3i$, with modulus $\sqrt{(-2)^2 + (-3)^2} = \sqrt{13}$.

(a) The point $(4, 1)$ corresponds to $4 + i$.; (b) The point $(-2, -3)$ corresponds to $-2 - 3i$, with modulus $\sqrt{(-2)^2 + (-3)^2} = \sqrt{13}$.

5 **[Foundational]** • **Target: Solving equations with complex roots**

Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.

By the quadratic formula: $x = \frac{-2 \pm \sqrt{4 - 20}}{2} = \frac{-2 \pm \sqrt{-16}}{2} = \frac{-2 \pm 4i}{2} = -1 \pm 2i$.

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6 [Foundational] • Target: Solving equations with complex roots

One root of a quadratic equation with real coefficients is $3 - 4i$. State the other root, and find the quadratic equation.

Since the coefficients are real, complex roots occur in conjugate pairs, so the other root is $3 + 4i$. Sum of roots $= 6$; product $= (3 - 4i)(3 + 4i) = 9 + 16 = 25$. Equation: $x^2 - 6x + 25 = 0$.

7 [Foundational] • Target: Solving equations with complex roots

Given that $z = 1 + 2i$ is a root of $z^3 - 4z^2 + 9z - 10 = 0$ (with real coefficients), find the other two roots.

Since the coefficients are real, $z = 1 - 2i$ is also a root. These two roots multiply to $(1 + 2i)(1 - 2i) = 1 + 4 = 5$ and sum to 2, giving the quadratic factor $z^2 - 2z + 5$. Dividing the cubic by this factor: $z^3 - 4z^2 + 9z - 10 \div (z^2 - 2z + 5) = z - 2$, so the third root is $z = 2$. The other two roots are $1 - 2i$ and 2.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Converting to polar (modulus-argument) form

Write $z = \sqrt{3} + i$ in polar form $r(\cos\theta + i\sin\theta)$, giving the argument in radians.

$r = \sqrt{(\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$. $\theta = \arctan\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6}$. Polar form: $z = 2\left(\cos\frac{\pi}{6} + i\sin\frac{\pi}{6}\right)$.

9 [Intermediate] • Target: Converting to polar (modulus-argument) form

Convert $z = 6\left(\cos\frac{2\pi}{3} + i\sin\frac{2\pi}{3}\right)$ to Cartesian form.

$a = 6\cos\left(\frac{2\pi}{3}\right) = 6\left(-\frac{1}{2}\right) = -3$. $b = 6\sin\left(\frac{2\pi}{3}\right) = 6\left(\frac{\sqrt{3}}{2}\right) = 3\sqrt{3}$. So $z = -3 + 3\sqrt{3}i$.

10 [Intermediate] • Target: Multiplying and dividing in polar form

Given $z_1 = 4\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$ and $z_2 = 2\left(\cos\frac{\pi}{12} + i\sin\frac{\pi}{12}\right)$, find $z_1 z_2$ and $\frac{z_1}{z_2}$ in polar form.

Multiply moduli, add arguments: $z_1 z_2 = 8\left(\cos\frac{5\pi}{12} + i\sin\frac{5\pi}{12}\right)$. Divide moduli, subtract arguments: $\frac{z_1}{z_2} = 2\left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)$.

11 [Intermediate] • Target: Multiplying and dividing in polar form

Find the modulus and argument of $z = \frac{(1+i)^3}{1-i}$, using De Moivre's theorem for the numerator and polar division.

$1 + i$ has modulus $\sqrt{2}$ and argument $\frac{\pi}{4}$, so $(1 + i)^3$ has modulus $2\sqrt{2}$ and argument $\frac{3\pi}{4}$. $1 - i$ has modulus $\sqrt{2}$ and argument $-\frac{\pi}{4}$. Dividing: modulus $= \frac{2\sqrt{2}}{\sqrt{2}} = 2$; argument $= \frac{3\pi}{4} - \left(-\frac{\pi}{4}\right) = \pi$. (This means $z = -2$, purely real.)

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12 [Intermediate] • Target: De Moivre's theorem

Use De Moivre's theorem to find $\left[3\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)\right]^4$, giving your answer in Cartesian form.

$3^4\left(\cos\frac{4\pi}{6}+i\sin\frac{4\pi}{6}\right)=81\left(\cos\frac{2\pi}{3}+i\sin\frac{2\pi}{3}\right)=81\left(-\frac{1}{2}+\frac{\sqrt{3}}{2}i\right)=-\frac{81}{2}+\frac{81\sqrt{3}}{2}i$.

13 [Intermediate] • Target: Finding roots using De Moivre's theorem

Find the three cube roots of $z=27\left(\cos\frac{\pi}{2}+i\sin\frac{\pi}{2}\right)$, giving arguments in $[0,2\pi)$.

Cube roots have modulus $27^{1/3}=3$ and arguments $\frac{\pi/2+2\pi k}{3}$ for $k=0,1,2$: $\frac{\pi}{6}$, $\frac{5\pi}{6}$, $\frac{3\pi}{2}$. Roots: $3\left(\cos\frac{\pi}{6}+i\sin\frac{\pi}{6}\right)$, $3\left(\cos\frac{5\pi}{6}+i\sin\frac{5\pi}{6}\right)$, $3\left(\cos\frac{3\pi}{2}+i\sin\frac{3\pi}{2}\right)$.

14 [Intermediate] • Target: Finding roots using De Moivre's theorem

Find all four fourth roots of -16 (i.e. solve $z^4=-16$), giving arguments in $[0,2\pi)$.

Write $-16=16(\cos\pi+i\sin\pi)$. Fourth roots have modulus $16^{1/4}=2$ and arguments $\frac{\pi+2\pi k}{4}$ for $k=0,1,2,3$: $\frac{\pi}{4}$, $\frac{3\pi}{4}$, $\frac{5\pi}{4}$, $\frac{7\pi}{4}$. Roots: $2\left(\cos\frac{\pi}{4}+i\sin\frac{\pi}{4}\right)$, $2\left(\cos\frac{3\pi}{4}+i\sin\frac{3\pi}{4}\right)$, $2\left(\cos\frac{5\pi}{4}+i\sin\frac{5\pi}{4}\right)$, $2\left(\cos\frac{7\pi}{4}+i\sin\frac{7\pi}{4}\right)$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 1.01 Functions: Modulus & Inverse

This question has three parts. An AC circuit has impedance $Z=R+iX_L$, where $R=8\,\Omega$ (resistance) and $X_L=6\,\Omega$ (inductive reactance). (a) Write Z in Cartesian form, find $|Z|$, and find $\arg Z$ to the nearest degree. (b) The current is $I=\frac{V}{Z}$, where $V=100$ volts (real). Find I in the form $a+bi$, by rationalizing the denominator. (c) Find $|I|$, to two decimal places.

(a) $Z=8+6i$. $|Z|=\sqrt{64+36}=\sqrt{100}=10\,\Omega$. $\arg Z=\arctan\left(\frac{6}{8}\right)\approx 37^\circ$. (b) $I=\frac{100}{8+6i}\cdot\frac{8-6i}{8-6i}=\frac{100(8-6i)}{64+36}=\frac{100(8-6i)}{100}=8-6i$ amps. (c) $|I|=\sqrt{8^2+(-6)^2}=\sqrt{64+36}=\sqrt{100}=10.00$ amps, consistent with $|I|=\frac{|V|}{|Z|}=\frac{100}{10}=10$.

16 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 10.01 Numerical Methods

Convert the Cartesian equation $(x^2+y^2)^2=4(x^2-y^2)$ into polar form and identify the name and key features of the resulting curve.

Substitute standard Cartesian-to-polar relations $x^2+y^2=r^2$, $x=r\cos\theta$, and $y=r\sin\theta$: $(r^2)^2=4(r^2\cos^2\theta-r^2\sin^2\theta)\Rightarrow r^4=4r^2(\cos^2\theta-\sin^2\theta)$. Apply the double-angle identity $\cos^2\theta-\sin^2\theta=\cos(2\theta)$: $r^4=4r^2\cos(2\theta)$. Divide through by r^2 (for $req0$): $r^2=4\cos(2\theta)$. This is a lemniscate (figure-eight curve) centered at the pole with loops oriented along the polar axis and maximum reach $r=2$ at.

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17 **[Challenging]** • **Target: Finding roots using De Moivre's theorem** • **Cross-Lesson: 11.01 Differential Equations**

[Polar Coordinates, Complex Numbers & De Moivre's Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

18 **[Challenging]** • **Target: A well-designed word problem: AC circuit impedance** • **Cross-Lesson: 12.01 Poisson Distribution**

Find all five solutions to $z^5 = -32$ in polar form $re^{i\theta}$ with $-\pi < \theta \le \pi$. Show that the sum of these five roots equals 0.

1. Write $-32 = 32e^{i\pi}$. By De Moivre's Theorem: $z_k = 32^{1/5}\exp\left(i\frac{\pi + 2\pi k}{5}\right) = 2e^{i(\pi + 2\pi k)/5}$ for $k = 0, 1, 2, 3, 4$. Evaluating the arguments in $(-\pi, \pi]$: - $k = 0$: $z_0 = 2e^{i\pi/5}$ - $k = 1$: $z_1 = 2e^{i3\pi/5}$ - $k = 2$: $z_2 = 2e^{i5\pi/5} = 2e^{i\pi} = -2$ - $k = 3$: $z_3 = 2e^{i7\pi/5} = 2e^{-i3\pi/5}$ - $k = 4$: $z_4 = 2e^{i9\pi/5} = 2e^{-i\pi/5}$ 2. Sum of the roots: Notice that.

19 **[Challenging]** • **Target: Basic arithmetic with complex numbers** • **Cross-Lesson: 13.01 Continuous Random Variables**

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

Take the natural logarithm of both sides: $\ln(3^{2x+1}) = \ln(5^{x-2})$. Apply the power law of logarithms: $(2x + 1)\ln 3 = (x - 2)\ln 5$. Distribute: $2x\ln 3 + \ln 3 = x\ln 5 - 2\ln 5$. Collect all terms containing x on one side: $2x\ln 3 - x\ln 5 = -\ln 3 - 2\ln 5 \Rightarrow x(2\ln 3 - \ln 5) = -(\ln 3 + 2\ln 5)$. Simplify logarithmic coefficients: $2\ln 3 = \ln 9$ and $2\ln 5 = \ln 25$: $x(\ln 9 - \ln 5) = -(\ln 3 + \ln 25) \Rightarrow x\ln\left(\frac{9}{5}\right) = -\ln(75)$. Solve for x.

20 **[Challenging]** • **Target: The complex conjugate and division** • **Cross-Lesson: 14.01 Sampling & Estimation**

Express $\frac{3+\sqrt{5}}{2-\sqrt{5}}$ in the form $p + q\sqrt{5}$, where p and q are rational numbers to be found.

Multiply numerator and denominator by the conjugate of the denominator, $2 + \sqrt{5}$: $\frac{3+\sqrt{5}}{2-\sqrt{5}} \times \frac{2+\sqrt{5}}{2+\sqrt{5}} = \frac{(3+\sqrt{5})(2+\sqrt{5})}{2^2 - (\sqrt{5})^2}$ Expanding the numerator: $3(2) + 3\sqrt{5} + 2\sqrt{5} + (\sqrt{5})^2 = 6 + 5\sqrt{5} + 5 = 11 + 5\sqrt{5}$. The denominator is $4 - 5 = -1$. Therefore: $\frac{11+5\sqrt{5}}{-1} = -11 - 5\sqrt{5}$ Thus $p = -11$ and $q = -5$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Basic arithmetic with complex numbers

A complex number z is represented in polar form as $z = r(\cos(\theta) + i\sin(\theta))$. If a complex number has a modulus $r = 5$ and an argument $\theta = \frac{\pi}{3}$, what is its rectangular form $a + bi$?

a. $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$ ✓

b. $5 + i5$

c. $\frac{5\sqrt{3}}{2} + i\frac{5}{2}$

d. $5i$

The rectangular form of a complex number given in polar form $z = r(\cos(\theta) + i\sin(\theta))$ is $a + bi$, where $a = r\cos(\theta)$ and $b = r\sin(\theta)$. Given $r = 5$ and $\theta = \frac{\pi}{3}$: $a = 5\cos(\frac{\pi}{3}) = 5 \times \frac{1}{2} = \frac{5}{2}$
$b = 5\sin(\frac{\pi}{3}) = 5 \times \frac{\sqrt{3}}{2} = \frac{5\sqrt{3}}{2}$ Therefore, the rectangular form is $\frac{5}{2} + i\frac{5\sqrt{3}}{2}$.

22 [Foundational] • Target: The complex conjugate and division

A complex number z is represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Given that $z^3 = 8i$, and z lies in the first quadrant such that its argument θ satisfies $0 < \theta < \frac{\pi}{2}$. Find the value of $r^2\cos\theta$.

a. 2

b. **4** ✓

c. $2\sqrt{2}$

d. 1

We are given $z^3 = 8i$. In polar form, $8i = 8(\cos\frac{\pi}{2} + i\sin\frac{\pi}{2})$. Let $z = r(\cos\theta + i\sin\theta)$. Then $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. Equating the magnitudes: $r^3 = 8$, so $r = 2$. Equating the arguments: $3\theta = \frac{\pi}{2} + 2k\pi$ for integer k. Since z is in the first quadrant, $0 < \theta < \frac{\pi}{2}$. For $k = 0$, $3\theta = \frac{\pi}{2}$, so $\theta = \frac{\pi}{6}$. This lies in the first quadrant. For $k = 1$.

23 [Foundational] • Target: Modulus and the Argand diagram

Let the complex number z be represented in polar form as $z = r(\cos\theta + i\sin\theta)$. Suppose z satisfies the equation $z^3 = -64i$. If z is restricted to the fourth quadrant (i.e., $\frac{3\pi}{2} < \arg(z) < 2\pi$), find the real part of z.

a. 4

b. -4

c. $2\sqrt{3}$ ✓

d. $-2\sqrt{3}$

We are given the equation $z^3 = -64i$. First, express $-64i$ in polar form. The modulus is $|-64i| = 64$. The argument is $\arg(-64i) = \frac{3\pi}{2}$ (or $-\frac{\pi}{2}$). So, $-64i = 64\left(\cos\frac{3\pi}{2} + i\sin\frac{3\pi}{2}\right)$. Let $z = r(\cos\phi + i\sin\phi)$. Then $z^3 = r^3(\cos(3\phi) + i\sin(3\phi))$. Equating the polar forms, we have $r^3 = 64$ and $3\phi = \frac{3\pi}{2} + 2k\pi$ for some integer k. From $r^3 = 64$, we get $r = 4$ (since r is a non-negative real number). Fro.

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24 [Foundational] • Target: Solving equations with complex roots

Let the complex number z_1 be represented in polar form as $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$, where $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. Let another complex number, z_2, be represented as $z_2 = 2(\cos\theta_2 + i\sin\theta_2)$. If $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$ and $\alpha = \frac{5\pi}{6}$, find the value of θ_2 in radians, given that $0 \le \theta_2 < 2\pi$.

a. $\frac{\pi}{2}$ ✓

b. $\frac{2\pi}{3}$

c. $\frac{5\pi}{6}$

d. $\frac{\pi}{6}$

We are given two complex numbers in polar form: $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and $z_2 = r_2(\cos\theta_2 + i\sin\theta_2)$. We are provided with $r_1 = 4$ and $\theta_1 = \frac{\pi}{3}$. For z_2, we are given $r_2 = 2$. The product of two complex numbers in polar form is given by $z_1 \cdot z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$. We are given that $z_1 \cdot z_2 = 8(\cos\alpha + i\sin\alpha)$, where $\alpha = \frac{5\pi}{6}$. Equating the magnitudes, we.

25 [Foundational] • Target: Converting to polar (modulus-argument) form

Consider the complex number $z = \left(\frac{\sqrt{3}}{2} + \frac{1}{2}i\right)^{10}$. If z is expressed in polar form as $r(\cos\theta + i\sin\theta)$, where $r > 0$ and $0 \le \theta < 2\pi$, find the exact values of r and θ.

a. $r = 1, \theta = \frac{5\pi}{3}$ ✓

b. $r = 1, \theta = \frac{4\pi}{3}$

c. $r = 10, \theta = \frac{5\pi}{3}$

d. $r = \sqrt{3}, \theta = \frac{4\pi}{3}$

First, express the base complex number in polar form. Let $w = \frac{\sqrt{3}}{2} + \frac{1}{2}i$. The modulus of w is $|w| = \sqrt{\left(\frac{\sqrt{3}}{2}\right)^2 + \left(\frac{1}{2}\right)^2} = \sqrt{\frac{3}{4} + \frac{1}{4}} = \sqrt{1} = 1$. The argument of w is $\arg(w) = \arctan\left(\frac{1/2}{\sqrt{3}/2}\right) = \arctan\left(\frac{1}{\sqrt{3}}\right)$. Since the real and imaginary parts are both positive, w is in the first quadrant, so $\arg(w) = \frac{\pi}{6}$. Thus.

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26 [Foundational] • Target: Multiplying and dividing in polar form

Consider the complex number $z = r(\cos\theta + i\sin\theta)$. If $z^3 = -8$ and $r > 0$, and the principal argument of z is $\frac{\pi}{3}$, find the value of r.

a. **2** ✓

b. 4

c. 8

d. 16

We are given the complex number $z = r(\cos\theta + i\sin\theta)$, where $r > 0$. We are also given that $z^3 = -8$ and the principal argument of z is $\theta = \frac{\pi}{3}$. Using De Moivre's Theorem, we have $z^3 = r^3(\cos(3\theta) + i\sin(3\theta))$. We are given $z^3 = -8$, which can be written in polar form as $8(\cos(\pi + 2k\pi) + i\sin(\pi + 2k\pi))$ for any integer k. Equating the magnitudes, we get $r^3 = 8$. Since $r > 0$, we have $r = \sqrt[3]{8} = 2$. Equating the arguments, we get.

27 [Foundational] • Target: De Moivre's theorem

A complex number is represented by the point $P(3, 4)$ in the complex plane. What is the modulus of this complex number?

a. **5** ✓

b. 7

c. 25

d. 16

The complex number can be written as $z = 3 + 4i$. The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. In this case, $a = 3$ and $b = 4$. Therefore, the modulus is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

28 [Foundational] • Target: Finding roots using De Moivre's theorem

How many distinct complex nth roots does the equation $z^5 = 1$ have, and what is the angle (in degrees) between consecutive roots on the Argand diagram?

a. 5 roots, 72° apart ✓

b. 4 roots, 72° apart

c. 5 roots, 60° apart

d. 6 roots, 72° apart

$z^5 = 1$ has exactly 5 roots, equally spaced by $\frac{360^\circ}{5} = 72^\circ$.

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: A well-designed word problem: AC circuit impedance**

Consider the complex number $w = 4 - 3i$. Find the modulus of w, denoted as $|w|$.

a. **5** ✓

b. 6

c. 7

d. 25

The modulus of a complex number $z = x + yi$ is given by $|z| = \sqrt{x^2 + y^2}$. For the complex number $w = 4 - 3i$, we have $x = 4$ and $y = -3$. Therefore, $|w| = \sqrt{4^2 + (-3)^2} = \sqrt{16 + 9} = \sqrt{25} = 5$. Common errors include: - Forgetting to square the imaginary part: $\sqrt{4 + (-3)} = \sqrt{1} = 1$ (Incorrect). - Forgetting to take the square root: $4^2 + (-3)^2 = 16 + 9 = 25$ (Incorrect, as this is the square of the modulus). - Incorrectly handling the negative sign in the imaginary part.

30 **[Intermediate]** • **Target: Basic arithmetic with complex numbers**

Let $z = x + yi$ be a complex number. The modulus of z, denoted by $|z|$, is given by $|z| = \sqrt{x^2 + y^2}$. Given that $|z| = \sqrt{5}$ and the real part of z is 2, find the possible values for the imaginary part of z.

a. $\pm\sqrt{3}$ ✓

b. ± 3

c. $\pm\sqrt{1}$

d. ± 1

We are given that $z = x + yi$. The modulus of z is $|z| = \sqrt{x^2 + y^2}$. We are given that $|z| = \sqrt{5}$, so $\sqrt{x^2 + y^2} = \sqrt{5}$. Squaring both sides, we get $x^2 + y^2 = 5$. We are also given that the real part of z is 2, which means $x = 2$. Substituting $x = 2$ into the equation $x^2 + y^2 = 5$, we get $2^2 + y^2 = 5$, which simplifies to $4 + y^2 = 5$. Subtracting 4 from both sides, we have $y^2 = 1$. Taking the square root of both sides, we find $y = \pm 1$. Therefore, the possible valu.

31 **[Intermediate]** • **Target: The complex conjugate and division**

Consider the complex number $z = 3 + 4i$. What is the modulus of z?

a. **5** ✓

b. 7

c. 25

d. $\sqrt{7}$

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For the given complex number $z = 3 + 4i$, we have $a = 3$ and $b = 4$. Therefore, the modulus of z is $|z| = \sqrt{3^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$. Common misconceptions: - Adding the real and imaginary parts: $3 + 4 = 7$, which is incorrect. - Squaring the real and imaginary parts without taking the square root: $3^2 + 4^2 = 9 + 16 = 25$, which is the square of the modulus. - Incorrectly calculating the square root.

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32 [Intermediate] • Target: Modulus and the Argand diagram

Consider the complex number $z = 3 + ki$, where k is a real parameter. The modulus of z is given by $|z| = \sqrt{13}$. Find the positive value of k.

a. **1 ✓**

b. 2

c. 3

d. 4

The modulus of a complex number $z = a + bi$ is given by $|z| = \sqrt{a^2 + b^2}$. For the complex number $z = 3 + ki$, we have $a = 3$ and $b = k$. Therefore, $|z| = \sqrt{3^2 + k^2} = \sqrt{9 + k^2}$. We are given that $|z| = \sqrt{13}$. So, we have the equation: $\sqrt{9 + k^2} = \sqrt{13}$ Squaring both sides of the equation, we get: $9 + k^2 = 13$ Subtracting 9 from both sides: $k^2 = 13 - 9$ $k^2 = 4$ Taking the square root of both sides, we get $k = \pm 2$. Since we are asked for the positive value of k, w.

33 [Intermediate] • Target: Solving equations with complex roots

Consider the complex numbers $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. Find the value of $z_1 - z_2$ in the form $x + yi$.

a. **4 - 6i ✓**

b. 6 + 6i

c. 4 + 6i

d. 6 - 6i

We are asked to find the difference between two complex numbers, z_1 and z_2. Given $z_1 = 5 - 2i$ and $z_2 = 1 + 4i$. We need to calculate $z_1 - z_2$. $z_1 - z_2 = (5 - 2i) - (1 + 4i)$ To subtract complex numbers, we subtract the real parts and the imaginary parts separately. Real part: $5 - 1 = 4$ Imaginary part: $(-2i) - (4i) = -2i - 4i = -6i$ Therefore, $z_1 - z_2 = 4 - 6i$. This matches choice A.

34 [Intermediate] • Target: Converting to polar (modulus-argument) form

Consider the complex number $z = -3 + 4i$. Find the modulus of z, denoted by $|z|$.

a. **5 ✓**

b. 7

c. 25

d. 16

The modulus of a complex number $z = a + bi$ is given by the formula $|z| = \sqrt{a^2 + b^2}$. In this case, $a = -3$ and $b = 4$. Therefore, $|z| = \sqrt{(-3)^2 + 4^2} = \sqrt{9 + 16} = \sqrt{25} = 5$.

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35 [Intermediate] • Target: Multiplying and dividing in polar form

Using De Moivre's theorem, evaluate $(1+i)^9$.

a. $16-16i$

b. $16+16i$ ✓

c. $-16-16i$

d. $-16+16i$

$1+i=\sqrt{2}\left(\cos(\frac{\pi}{4})+i\sin(\frac{\pi}{4})\right)$, so the power is $16\sqrt{2}\left(\cos(\frac{9\pi}{4})+i\sin(\frac{9\pi}{4})\right)=16+16i$.

36 [Intermediate] • Target: De Moivre's theorem

Product of roots of $2x^2-5x+3=0$:

a. $\frac{3}{2}$ ✓

b. $\frac{5}{2}$

c. $-\frac{3}{2}$

d. 3

Applying Algebra, Proof & Complex Numbers principles, Product = c/a = 3/2. Therefore, the verified correct option is $\frac{3}{2}$.

37 [Intermediate] • Target: Finding roots using De Moivre's theorem

Find all cube roots of unity.

a. $1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$ ✓

b. $1, i, -1$

c. $1, -1, i$

d. $1, i\sqrt{3}, -i\sqrt{3}$

$z^3=1\Rightarrow z_k=e^{2\pi ik/3}$ for $k=0,1,2$: $1,\ \frac{-1+i\sqrt{3}}{2},\ \frac{-1-i\sqrt{3}}{2}$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: A well-designed word problem: AC circuit impedance • Cross-Lesson: 12.01 Poisson Distribution

For $x^2 - x - 56 = 0$, find the product of the roots.

a. -42

b. -47

c. **-56** ✓

d. -51

Working through this complex-numbers step: Product of roots $= \frac{c}{a} = -56 = -56$

39 [Challenging] • Target: Basic arithmetic with complex numbers • Cross-Lesson: 13.01 Continuous Random Variables

Simplify $(1 + 2i)(5 + 5i)$, giving your answer in the form $x + yi$.

a. $-3 + 16i$

b. $-5 + 15i$ ✓

c. $-1 + 12i$

d. $-3 + 14i$

Working through this complex-numbers step: $(1 + 2i)(5 + 5i) = (5) + (5)i + 2(5)i + 2(5)i^2 = -5 + 15i$

40 [Challenging] • Target: The complex conjugate and division • Cross-Lesson: 14.01 Sampling & Estimation

For $x^3 - 19x - 30 = 0$, find the product of the three roots.

a. **30** ✓

b. 43

c. 27

d. 20

Working through this complex-numbers step: Product of roots $= -\frac{d}{a} = -(-30) = 30$

41 [Challenging] • Target: Modulus and the Argand diagram • Cross-Lesson: 15.01 Hypothesis Testing

Write $z = 4(\cos 180^\circ + i\sin 180^\circ)$ in the form $x + yi$ (exact values).

a. $-4 + 1$

b. 4

c. -4 ✓

d. $-4i$

Working through this complex-numbers step: $z = 4\cos 180^\circ + 4\sin 180^\circ\, i = -4$

42 [Challenging] • Target: Solving equations with complex roots • Cross-Lesson: 2.01 Exponentials & Logarithms

Parametric area using Green's theorem: $A = \frac{1}{2}\oint(x\,dy - y\,dx)$. For ellipse $x = a\cos t, y = b\sin t$:

a. $\pi ab/2$

b. πab ✓

c. $2\pi ab$

d. ab

$A = \frac{1}{2}\int_0^{2\pi}(a\cos t \cdot b\cos t - b\sin t \cdot (-a\sin t))dt = \pi ab$.

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43 [Challenging] • Target: Converting to polar (modulus-argument) form • Cross-Lesson: 3.01 Advanced Trigonometry

Find $\bar{z}$ (the complex conjugate) if $z = -3 + 9i$.

a. $-3 - 9i$ ✓

b. $-3 + 9i$

c. $3 - 9i$

d. $3 + 9i$

The conjugate negates the imaginary part: $\bar{z} = -3 - 9i$

44 [Challenging] • Target: Multiplying and dividing in polar form • Cross-Lesson: 4.01 Differentiation

The diagram shows the graphs of two linear equations. The point of intersection of the two lines represents the solution to the system of simultaneous equations. Find the coordinates of the point of intersection.

a. (1, 2)

b. **(2, 1)** ✓

c. (3, 0)

d. (0, 3)

The point of intersection of two lines is the solution to the system of simultaneous equations that represent these lines. By visually inspecting the graph, we can identify the coordinates where the two lines cross. The first line appears to pass through the points (0, 3) and (3, 0). The second line appears to pass through the points (0, 1) and (2, 3). Let's find the equations of the lines. For the first line: Gradient $m_1 = \frac{0-3}{3-0} = \frac{-3}{3} = -1$. Using the point-slope form with (3, 0).

45 [Challenging] • Target: De Moivre's theorem • Cross-Lesson: 5.01 Integration

The diagram shows the graph of the function $y = f(x)$. The function is of the form $f(x) = x^2 + bx + c$. The graph passes through the point $(2, -3)$ and has a minimum value of -4 at $x = 1$. Find the value of b.

a. -2

b. 1

c. **2** ✓

d. -1

The function is given by $f(x) = x^2 + bx + c$. The minimum value of a quadratic function in the form $ax^2 + bx + c$ occurs at $x = -\frac{b}{2a}$. In this case, $a = 1$, so the minimum occurs at $x = -\frac{b}{2}$. We are given that the minimum occurs at $x = 1$. Therefore, $1 = -\frac{b}{2}$, which implies $b = -2$. Alternatively, we know the vertex is at $(1, -4)$. The vertex form of a quadratic is $f(x) = a(x - h)^2 + k$. Given $a = 1$ and the vertex $(h, k) = (1, -4)$, we have $f(x) = 1(x - 1)^2 - 4$. Expanding thi.

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